Carleton University | Faculty of Engineering and Design

MECH 5800 - Advanced Optimization Modelling

TITLE WAHOO

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1 Problem Introduction

Public transport is an important part of any metropolitan city. It provides essential mobility to people without the ability to drive or access to a car.

2 Methods

2.1 Model Formulation

2.1.1 Model Approach

The optimization of frequent transit routes is based on an economic framework, specifically a cost–benefit analysis that incorporates non-market valuation methods to quantify the social benefits associated with changes in bus route frequency. The optimal system is defined as one that allocates the available bus fleet across routes in a manner that maximizes overall socio-economic benefits. This leads to an optimization problem in which an integer number of buses must be assigned to each route.

The current transit system serves as the baseline against which improvements are evaluated. It is characterized using existing data on passenger demand for each route, along with current service frequencies obtained from published data by OC Transpo. As a result of the only available ridership data being aggregated over time blocks of the day as described in Table 1, the choice of assuming constant route frequency over a block of time is done. In other words for a given bus route there will be an optimal number of buses for a given time block. A more detailed description of the data sources and parameters used is provided in Section 2.3.

Table 1: Time blocks

Time Block	Time Period
Early AM	3:00-6:30
AM Peak	6:30-9:00
Midday	9:00-15:00
PM Peak	15:00-18:30
Evening	18:30-23:00
Night	23:00-3:00

The model of this problem will result in a multi-integer programming (MIP) problem, with the decision variable being the number of buses assigned to a route. In attempt to improve the efficiency of the MIP solver, cutting plane techniques will used and their efficacy will be investigated.

2.1.2 Evaluating Public Transport Benefits and Costs

There exists substantial academic literature on methods for evaluating the costs and benefits of non-market impacts in project analysis. Public sector projects, in particular, rely heavily on these methods, as many benefits related to the public good, public health, and the environment are not directly priced in markets. Public transportation is a clear example of a service that generates both direct and indirect benefits, including improved equity for individuals without access to a personal vehicle, enhanced public health outcomes due to increased walking to and from transit stops, reduced emissions per kilometre travelled, and decreased road congestion.

As discussed in the previous section, the proposed model requires quantifying changes in costs and benefits as a function of bus frequency on a given route. The level of detail and scope for quantifying numerically all possible benefits associated with a transport project can be extensive and not appropriate for the time constraints of this project. Therefore, the scope is restricted to three primary effects: the impact of increased bus frequency on ridership, the benefit of reduced passenger waiting times, and the environmental benefits associated with a reduction in personal vehicle use.

In general, the evaluation of these benefits is complex, inherently uncertain, and often context-specific. However, established best practices from the literature (e.g., [insert sources]) are used to guide the methodology. In contrast, costs are more straightforward to estimate, as increased service frequency directly leads to higher operating and maintenance expenses. The set of costs and benefits summarized in Table 2 is considered sufficient to capture the primary effects of frequency changes within the scope of this study.

Table 2: In scope benefits and costs of increased transit service frequency.

Improvement	Benefits	Costs
Increased Frequency	- Reduced wait time - Reduced CO2eq intensity per km travelled	- Increased maintenance costs - Increased fuel costs - Increased number of bus drivers

2.1.3 Cost and Benefit Equations

Before formulating the global optimization problem (i.e., the minimization of costs and maximization of benefits) the equations used to quantify these costs and benefits are first introduced. It is noted that in the following equations $n_{new,r,t}$ represents the number of buses in the optimized system while $n_{old,r,t}$ represents the number of current system, and the indices r and t denote the given route and time block, respectively.

The change in labour cost for a change in the number of buses for route r and time block t is calculated as:

$$\Delta C_{labour,r,t} = (n_{new,r,t} - n_{old,r,t}) \cdot H_{block,t} \cdot W_{driver} \tag{1}$$

where $H_{block,t}$ is the service hours within the given time bock and W_{driver} is the hourly wage rate of an OC Transpo bus driver.

The change in fuel cost for a change in the number of buses for route r and time block t is calculated as:

$$\Delta C_{fuel,r,t} = (n_{new,r,t} - n_{old,r,t}) \cdot L_r \cdot \frac{H_{block,t}}{\overline{T}_{r,t}} \cdot P_{fuel} \cdot FC_t \tag{2}$$

where L_r is the length of route, $\overline{T}_{r,t}$ is the average time of the route, P_{fuel} is the price of fuel per litre and FC is the fuel consumption of a bus in litres per km travelled.

Finally, the change in maintenance cost for a change in the number of buses for route r and time block time t is calculated as:

$$\Delta C_{maintenance,r,t} = (n_{new,r,t} - n_{old,r,t}) \cdot L_r \cdot \frac{H_{block,t}}{\overline{T}_{r,t}} \cdot P_{maintenance} \tag{3}$$

where $P_{maintenance}$ is the cost of maintenance per km travelled.

The demand (or ridership) is shown to change as a function of frequency. From **??** it is suggested that the elasticity of demand to frequency is 0.5 and therefore the new ridership $x_{new,r,t}$ can be calculated as:

$$x_{new,r,t}(f) = x_{old,r,t} \left(\frac{f_{new,r,t}}{f_{old,r,t}} \right)^{0.5} \tag{4}$$

In terms of number of buses instead the elasticity relationship can equally be written as:

$$x_{new,r,t}(n) = x_{old,r,t} \left(\frac{\frac{n_{new,r,t}}{\overline{T}_{r,t}}}{\frac{n_{old,r,t}}{\overline{T}_{r,t}}} \right)^{0.5} \tag{5}$$

$$= x_{old,r,t} \left(\frac{n_{new,r,t}}{n_{old,r,t}} \right)^{0.5} \tag{6}$$

The total benefit associated from the reduced waiting time for passengers due to a change in the number of buses for route r and time block time t is calculated as:

$$\Delta B_{time,r,t} = \frac{1}{2} \left(\frac{1}{f_{old,r,t}} - \frac{1}{f_{new,r,t}} \right) \cdot x_{new,r,t} \cdot W_{passenger} \cdot F_{wage,t} \tag{7}$$

$$= \frac{\overline{T}_{r,t}}{2} \left(\frac{1}{n_{old,r,t}} - \frac{1}{n_{new,r,t}} \right) \cdot x_{old,r,t} \left(\frac{n_{new,r,t}}{n_{old,r,t}} \right)^{0.5} \cdot W_{passenger} \cdot F_{wage,t} \tag{8}$$

where $W_{passenger}$ is the average wage and F_{wage} is the fraction of the hourly wage rate used to value the travel time saved of each passenger. The fraction of hourly wage will be higher in peak hours, due to the fact that during this time a higher percentage of people are commuting to work and therefore the value of time is higher as compared to other times, in addition to the fact travelling in higher congestion increases associated value of time travel []. Also, note that it is estimated that the average wait time of a passenger is

half of the headway time.

An effect of increasing the frequency of buses is a decrease in automobile use []. Newman-Kenworthy found that for 23 US, Canadian, Australian and European cities each transit passenger-mile represents a reduction of 3.6 automobile vehicle-miles (5.8 km). As a result of GHG emissions having a social cost we can calculate the benefit of avoided vehicle emissions as:

$$\Delta B_{emissions,r,t} = I_{GHG,car} \cdot V_{km,saved} \cdot (x_{new,r,t} - x_{old,r,t}) \cdot \overline{L}_{r,passenger} \cdot SCC \tag{9}$$

$$= I_{GHG,car} \cdot V_{km,saved} \cdot \left(x_{old,r,t} \left(\frac{n_{old,r,t}}{n_{new,r,t}} \right)^{0.5} - x_{old,r,t} \right) \cdot \overline{L}_{r,passenger} \cdot SCC \tag{10}$$

where $I_{GHG,car,t}$ is the greenhouse gas emission intensity per passenger km travelled in a car, $V_{km,saved}$ is the automobile vehicle-kilometres saved per 1 km of transit passenger km travelled, $\overline{L}_{r,passenger}$ is the average length of the passenger's commute and SCC is the social cost of carbon.

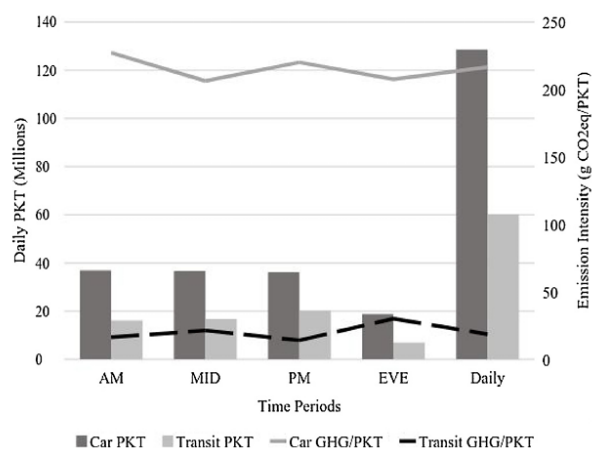

Figure 1: Passenger kilometres travelled (million PKT) in private vehicles and public transit and GHG emission intensity ($g\ CO_{2eq}/PKT$ []

The emission intensity values for buses and personal vehicles will depend on the time block, t. For transit the the emission intensity is affected by the volume of passengers, during off-peak hours buses will carry a lower passenger volume and therefore be less efficient in terms of emission per passenger. This effect can be seen in Figure 1 which shows that during the evening and mid day the emission intensity is higher for bus transit. For personal vehicles in peak hours congestion will cause higher emission intensities. The emission intensities for each time block is given in Table **??**.

Additionally, to approximate the distance a passenger travels we will use the simple approximation that it is 40% of the total bus route length, L_r. Again, it is highlighted that there is a fairly high level of uncertainty with many of the model parameters due to the lack of granular or primary data.

2.1.4 Linearization

For the ease of application of a linear integer solver the non-linear terms introduced in the benefit equations will be linearized. The selected method to linearize the equations is a first order Taylor expansion.

The first order Taylor expression around a point x_0 is defined as:

$$f(x) = f(x_0) + f'(x_o) \cdot (x - x_o) \tag{11}$$

In Equation 10 the non-linear terms is:

$$\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5}$$

The linear approximation is:

$$\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5} \approx 1 + \frac{0.5}{n_{old,r,t}}(n_{new,r,t} - n_{old,r,t}) \tag{12}$$

$$\tag{13}$$

Therefore the emissions cost benefit can be expressed as:

$$\Delta B_{emission,r,t} \approx I_{GHG,r,t} \cdot V_{km,saved} \cdot \overline{L}_{r,passenger} \cdot SCC \cdot \alpha \cdot (n_{new,r,t} - n_{old,r,t}) \tag{14}$$

Where $\alpha = \frac{0.5x_{old,r,t}}{n_{old,r,t}}$.

In Equation 8 the non-linear term is:

$$\left(\frac{1}{n_{old,r,t}} - \frac{1}{n_{new,r,t}}\right)\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5}$$

The linear approximation is:

$$\left(\frac{1}{n_{old,r,t}} - \frac{1}{n_{new,r,t}}\right)\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5} \approx \frac{1}{n_{old,r,t}^2}(n_{new,r,t} - n_{old,r,t}) \tag{15}$$

Therefore the value of time saved equation can be expressed as:

$$\Delta B_{time,r,t} \approx \frac{\overline{T}_{r,t}}{2} \cdot W_{passenger} \cdot F_{wage,t} \cdot \beta \cdot (n_{new,r,t} - n_{old,r,t}) \tag{16}$$

Where $\beta = \frac{x_{old,r,t}}{n_{old,r,t}^2}$

2.1.5 Constraints

2.2 Global Minimization Problem

$$\min_{n_{\text{new},r,t}} \quad \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}} \Delta\Big(C_{\text{labour},r,t} + C_{\text{maintenance},r,t} + C_{\text{fuel},r,t} - B_{\text{time},r,t} - B_{\text{emissions},r,t}\Big) \tag{17}$$

$$\text{s.t.} \quad \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}} \Big(C_{\text{labour},r,t} + C_{\text{maintenance},r,t} + C_{\text{fuel},r,t}\Big) \leq Budget_{\text{TotalOperational}} \tag{18}$$

$$\sum_{r\in\mathcal{R}} n_{\text{new},r,t} \leq Buses_{\text{tot}}, \quad \forall t \in \mathcal{T} \tag{19}$$

$$n_{r,t}^{\min} \leq n_{\text{new},r,t} \leq n_{r,t}^{\max}, \quad \forall r = 1,\ldots,R,\ \forall t \in \mathcal{T} \tag{20}$$

For the minimum number of buses we assume that the frequency must be at least once per hour in off peak times and twice per hour for on peak times.

2.3 Data Sources and Parameter Estimation

The two primary data sources used in this analysis were ridership data for a sample month in 2023 and current route data published by OC Transpo. On April 27, 2025, OC Transpo launched *New Ways to Bus*, a significant overhaul of the bus network. This initiative involved retiring some routes due to low ridership and improving connections to new routes and the O-Train. As a result of these updates, there is some mismatch between the ridership data used and the route data. Consequently, the model only optimizes bus routes for which 2023 ridership data are available. It is also acknowledged that these ridership trends may not fully reflect current patterns following the *New Ways to Bus* updates. Additional comments about the nature of each data source are as follows.

2.3.1 Ridership data

2.3.2 Route data

The data used to characterize the current system ($T_{r,t}$, $f_{old,r,t}$, $n_{old,r,t}$) was obtained from the General Transit Feed Specification (GTFS) provided by OC Transpo. GTFS is a standardized data format that public transit agencies publish, containing timing and geographic information on routes and stops. It includes both static data, which describes predetermined scheduled information, and real-time extension data, which provides live transit updates such as the current location of a bus for trip planning platforms. In this case, the static data is appropriate.

As the ridership data aggregates the total number of passengers during a time block for both outbound and

inbound routes, all route data is taken as an average of the two. Consequently, the average route length represents the mean of the outbound and inbound lengths during the given time block. While this is not always strictly accurate, it is a reasonable simplification for our purposes. The same principle applies to the average headway or frequency of a bus. At certain times of day, inbound service may be more frequent than outbound service; however, this is simplified to an average of the two due to the nature of the ridership data.

A Appendix

Table 3: Problem parameters; Empty fields are a work-in-progress.

Parameter	Description	Value	Units
$x_{new,r,t}$	New passenger demand as a result of frequency change		passengers/(route*time of day (TOD)
$x_{old,r,t}$	Current passenger demand		passengers/(route*TOD)
$f_{new,r,t}$	New service frequency		1/hr
$f_{old,r,t}$	Current service frequency		1/hr
$\overline{W}_{bus,driver}$	Average wage of bus driver		$/hr
$\overline{W}_{rider}$	Average hourly wage in Ottawa		$/hr
L_r	Length of route		km
$H_{service,r}$	Number of service hours of route		hr
FC	Fuel consumption of bus	22 [1]	litre/ 100 km
P_{fuel}	Average price of fuel	0.37 [2]	$/litre
$P_{maintenance}$	Average price of maintenance		$/vehicle km travelled
$I_{GHG,bus}$	GHG intensity of bus (diesel)	50 [3]	gCO2eq/passenger km travelled
$I_{GHG,car}$	GHG intensity of personal vehicle	290 [3]	gCO2eq/passenger km travelled
$\alpha_{diverted}$	Proportion of marginal transit riders diverted from personal vehicle use		-
F_{wage}	Fraction of hourly wage for value of travel time for bus passenger	0.35 [4]	-
SCC	Social cost of carbon	275 [5]	CAD$/tonne CO2eq

	Wkdy						
Route	Early AM	AM Peak	Midday	PM Peak	Evening	Night	**Wkdy**
1							**60600***
R2	43	1039	2231	1628	786	285	**6013**
5	23	292	566	303	162	73	**1419**
6	177	1724	3513	2781	1064	448	**9707**
7	110	1649	3533	2555	1075	550	**9472**

Figure 2: Excerp from route-level OC Transpo ridership data [6]

ELMVALE A	RING ROAD / TOH GENERAL CAMPUS	MAIN / HERRIDGE	ELGIN / MCLEOD	RIDEAU B	WALLER / NICHOLAS
05:22	05:27	05:33	05:37	05:44	05:49
05:59	06:04	06:11	06:15	06:22	06:28
06:15	06:20	06:27	06:31	06:38	06:44
06:30	06:36	06:43	06:47	06:54	06:59

Figure 3: Excerp from OC Transpo's "Schedules & Maps" webpage; Route 5 (Waller) [7]

References

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